

Computer Architecture
CSE 511/ECE 511
Mid-Semester (Monsoon 2020) Examination

DATE: 14 October, 2020

Instructions:

- The duration of the exam is **60 minutes**.
- The exam is for a total of **50 Marks**.
- **All questions** are compulsory.
- Please list down your assumptions (if any) clearly.
- Please make sure to properly differentiate different stages while writing pipeline diagrams.

Q1. Consider a processor with 5 pipeline stages. The delay of each pipeline stage are 4 ns, 5 ns, 10 ns, 10 ns and 7ns respectively. The registers residing between the pipeline stages has a delay of 2 ns. Determine the approximate speedup of the pipelined as compared to its non-pipelined implementation? 5 Marks

Answer:

Time taken to execute N instructions in non-pipelined implementation = $(4 + 5 + 10 + 10 + 7)N = 36N$ (1.5 Marks)

Time taken to execute N instructions in pipelined implementation = $(\max\{4,5,10,10,7\} + 2)N = 12N$. (1.5 Marks)

Speedup = $36N / 12N = 3$ (2 Marks)

Q2. Consider the information given in Table 1 regarding Machine A and Machine B. A program takes 50,000 instructions when run on Machine A. How many instructions would be needed to run the same program on Machine B, so that both the machines have same execution time for the program. 5 Marks

	CPI	Clock rate
Machine A	1.5	600MHz
Machine B	2.5	800MHz

Answer: $(\text{CPUTime})_A = (\text{Instruction count})_A * (\text{CPI})_A * (\text{Clock cycle Time})_A$
 $= (50,000) * (1.5) / (600 * 10^6) \text{ ns}$ (1.5 Marks)

$(\text{CPUTime})_B = (\text{Instruction count})_B * (\text{CPI})_B * (\text{Clock cycle Time})_B$
 $= (I)_B * (2.5) / (800 * 10^6) \text{ ns}$ (1.5 Marks)

$(\text{CPUTime})_A = (\text{CPUTime})_B$
Hence, by solving the equation, we have, $(I)_B = 40,000$ (2 Marks)

Q3. Consider a 32-bit machine with set-associative cache. The total cache size is 16KB and it is a 4-way set-associative cache. The length of memory location is 1 byte and the block size is 16 Bytes.

- a) Calculate the total number of sets present in the cache.

- b) Draw the 32-bit memory address showing the tag, index and offset bits.
 c) Determine the set number where the content of memory address 10FFCFF_(HEX) is mapped. Give explanation or steps. **(2 + 4 + 4) Marks**

Answer:

- a) $n=32$
 $M=16KB=2^4KB=2^{14}B$
 $block=B=16B=2^4B$
 $E=4=2^2$
 $M=S \times E \times B$
 $S=M \div (E \times B)=2^{14} \div (2^2 \times 2^4)=2^{14-6}=2^8=256 \text{ sets}$
 b) tag = 20
 Index = 8
 Offset = 4
 c) 207. You have to show the steps for full credit.

Q4. Consider a fully bypassed 5-stage RISC-V processor with the following pipeline stages: **F, D, X, M, W**. Whenever a branch instruction is encountered, it is resolved in the **E** Stage of the pipeline. The **F Stage** always speculates that the next PC is PC+4. Here, concentrate on only conditional branches and ignore any unconditional jumps. Consider the following set of instructions given below. Note that value of register R0 is 0.

0x1000: ADDI R4, R0, 9
 0x1004: LD R5, 0(R0)
 0x1008: ADD R5, R5, R4
 0x100c: BEQ R4, R5, 0x1000
 0x1010: LD R7, 4(R6)
 0x1014: ADD R5, R7, R5
 0x1018: SUB R1, R2, R3

Draw the pipeline timing diagram (resource usage diagram will not be considered) of the above instructions. You can draw the pipeline till seven instructions. Calculate how many clock cycles will be required to complete till seven instructions. **[15 Marks]**

Answer: 12 Clock cycles.

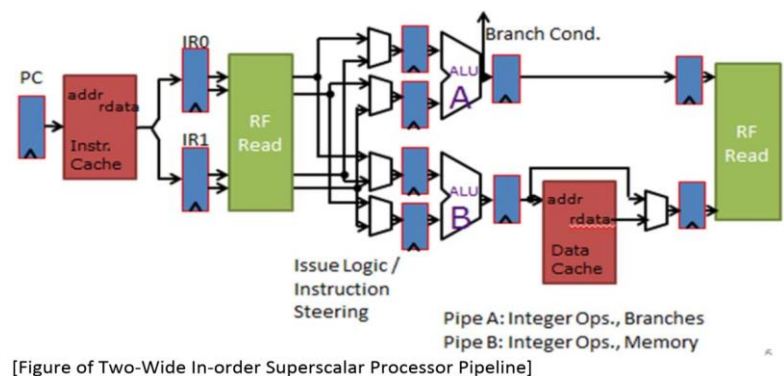
PC	Inst.	0	1	2	3	4	5	6	7	8	9	10	11
0X1000	ADDI	F	D	X	M	W							
0X1004	ADDI LD		F	D	X	M	W						
0X1008	ADD			F	D	D	X	M	W				
0X100C	BE8				F	F	D	X	M	W			
0X1010	LD					F	D	-	-	-			
0X1014	ADD						F	-	-	-			
0X1000	ADDI							F	D	X	M	W	

1+1+2+2+2+2+2= 12 FOR PIPELINE

3 FOR CLOCK CYCLES

Q5. Draw the pipeline diagram of the following code executing on the shown in-order two-way superscalar processor in the figure. Each pipeline stage takes one cycle. Assume that branches execute in pipeline A, loads/stores in pipeline B. Assume full bypassing and no alignment problems. Calculate how many clock cycles will be required to complete all the instructions. [15 Marks]

Inst1: ADD R14, R16, R17
 Inst2: SUB R1, R2, R3
 Inst3: SUB R2, R3, R5
 Inst4: LD R11, 0(R2)
 Inst5: ADDI R12, R13, 1
 Inst6: LD R15, 4(R2)
 Inst7: LD R15, 4(R15)
 Inst8: SUB R2, R9, R11
 Inst9: ADDI R3, R11, R12
 Inst10: ADD R4, R6, R7



Answer: Total 11 clock cycles.

	0	1	2	3	4	5	6	7	8	9	10
① ADD R14, R16, R17	F	D	A0	A1	W						
② SUB R1, R2, R3	F	D	B0	B1	W						
③ SUB R2, R3, R5		F	D	A0	A1	W					
④ LD R11, 0(R2)		F	D	D	B0	B1	W				
⑤ ADDI R12, R13, 1			F	D	A0	A1	W				
⑥ LD R15, 4(R2)			F	D	D	B0	B1	W			
⑦ LD R15, 4(R15)				F	D	D	D	B0	B1	W	
⑧ SUB R2, R9, R11				F	D	D	A0	A1	W		
⑨ ADD R3, R11, R12					F	D	D	A0	A1	W	
⑩ ADD R4, R6, R7					F	D	D	D	A0	A1	W

$$0.5+0.5+1+1.5+1+1.5+1.5+1.5+1.5+1.5 = 12$$

3 FOR CLOCK CYCLES

Explanation of stalls at different instructions:

Inst4: R2 available after A0 cycle.

Inst6: Decode hardware of pipeline B is used in Inst4

Inst7: R15 is available only after B1 stage of Inst6

Inst8: R11 is available after B1 stage of Inst4

Inst9: Decode hardware of A pipeline is used in Inst8.

Inst10: Decode stage of A pipeline is used in Inst9 and Decode stage of B pipeline is busy in Inst7.

